

DeepSeek-V2

Efficient MoE Language Model with Multi-Head Latent Attention

236B

Total Parameters

21B

Activated per Token

128K

Context Length

8.1T

Pretraining Tokens

The LLM Scaling Dilemma

Performance vs. Training Cost and Inference Speed

Prohibitive Training Cost

Dense 70B-class models require massive compute.

Training runs cost millions and scale quadratically with parameter count.

KV Cache Bottleneck

Standard MHA stores full K/V per head per token.

Context length caps inference at thousands of tokens before memory explodes.

Slow Inference Throughput

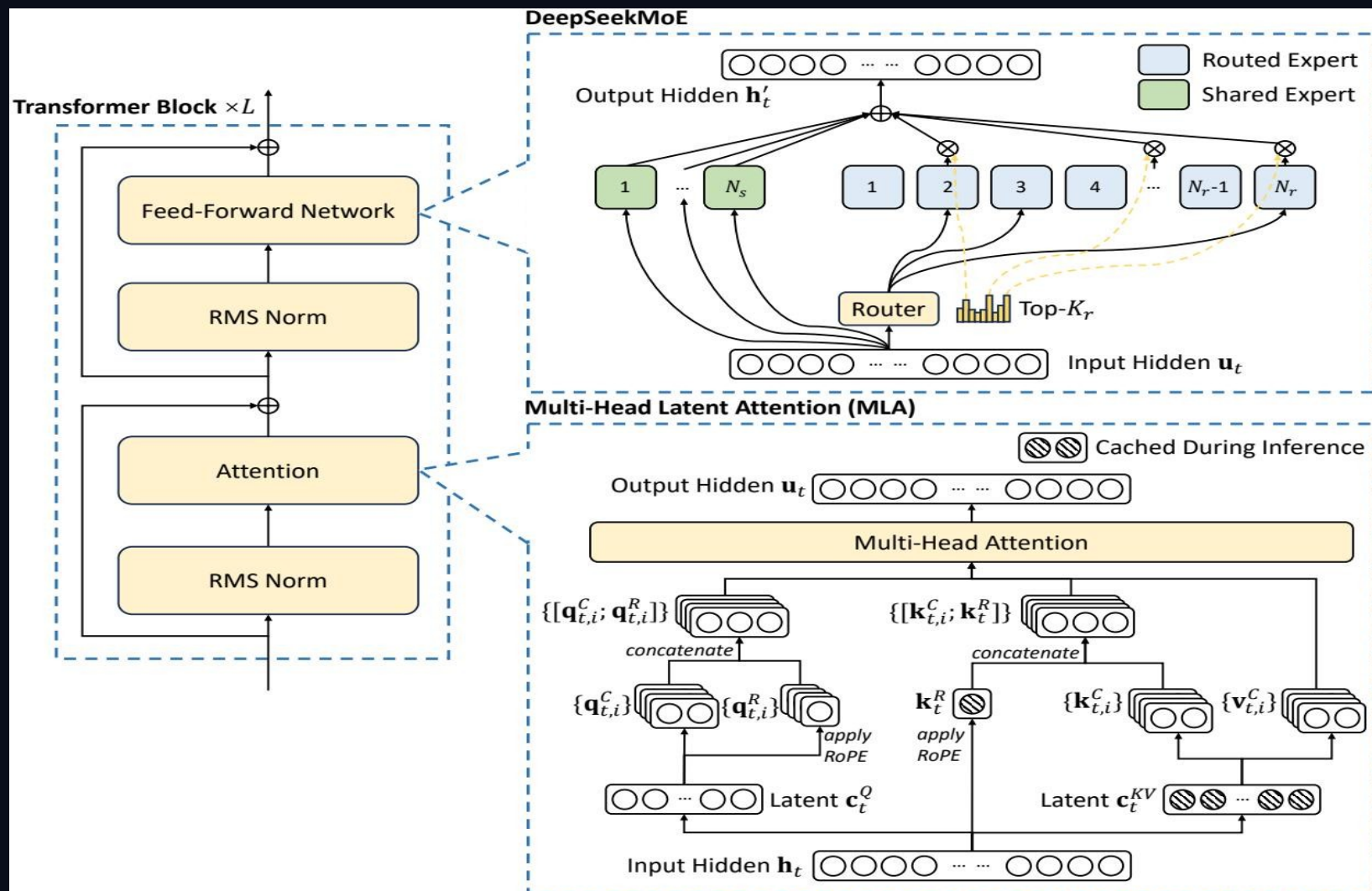
Dense activation means every forward pass touches

all 70B+ parameters, making deployment costly and latency unacceptable.

→ MoE + Efficient Attention: activate fewer parameters, compress state, preserve quality

DeepSeek-V2 Architecture

MLA + DeepSeekMoE in a Transformer Block

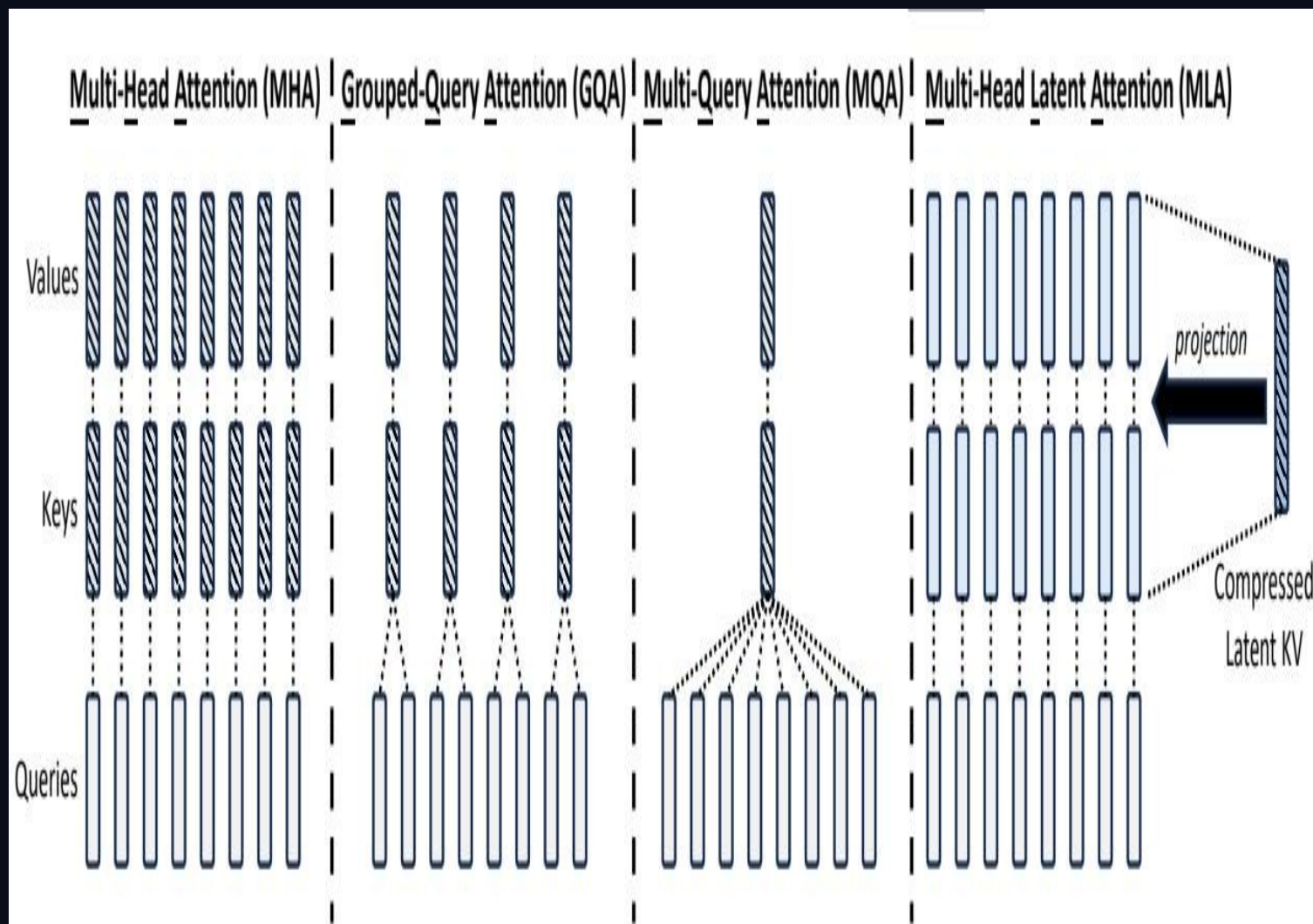


Key Facts

- MLA replaces standard MHA attention
- DeepSeekMoE replaces dense FFN
- Shared experts: always activated
- Routed experts: top-K sparse activation
- Device-limited routing $\rightarrow$ less comms
- Auxiliary load-balance losses
- Token-dropping during training
- Supports 128K context

Multi-Head Latent Attention (MLA)

Low-rank KV compression reduces cache by 93.3% — without sacrificing performance



How MLA Works

- MHA: full K/V per head → huge cache
- GQA/MQA: share K/V across heads → less cache, lower quality
- MLA: compresses K/V into a compact latent vector c_t^{KV}
- c_t^{KV} is cached; full K/V reconstructed on-the-fly via learned projections
- Decoupled RoPE-based positional key preserves positional info
- Result: better performance than MHA with 93.3% less KV cache

93.3% KV cache reduction

DeepSeekMoE

Fine-Grained Experts with Shared Expert Isolation

Expert Architecture

SHARED EXPERTS (N_s)

Always activated • Capture common knowledge

↓ Token

Router (top- K_r routing)

ROUTED EXPERTS (N_r)

Sparsely activated • Fine-grained specialization • Device-limited routing

Reduces cross-node communication • Auxiliary loss for load balance

Training Efficiency Features

► Fine-grained expert segmentation

Breaks FFN into smaller, specialized experts for richer activation patterns.

► Shared expert isolation

Shared experts always active → capture universal knowledge; routed experts handle specialization.

► Device-limited routing

Each token routed to experts on $\leq M$ devices — limits expensive cross-node communication.

► Auxiliary load-balancing losses

Prevent expert collapse; ensure balanced compute distribution across devices.

► Token-dropping strategy (training only)

Drops tokens when experts are overloaded — no effect on inference latency.

► 42.5% training cost reduction

vs. DeepSeek 67B dense model — enabled by sparse expert activation.

Efficiency Gains vs. DeepSeek 67B

DeepSeek-V2 delivers 42.5% training cost reduction, 93.3% KV cache reduction, and 5.76× throughput

DeepSeek 67B (Dense)			DeepSeek-V2 (MoE + MLA)
Training Cost	100%	—	57.5% (−42.5%)
KV Cache Size	100%	—	6.7% (−93.3%)
Throughput	1×	—	5.76×
Activated Params	67B	—	21B
Total Params	67B	—	236B

All metrics are relative comparisons vs. DeepSeek 67B. No absolute FLOPs or memory values fabricated.

MMLU Performance vs. Activated Parameters

DeepSeek-V2 achieves the best MMLU among open-source models at only 21B activated parameters

MMLU



Benchmark Results

Model	Act. B	MMLU
DeepSeek-V2	21	78.5
LLaMA 3 70B	70	77.0
Mixtral 8×22B	39	77.6
Qwen1.5 72B	72	75.6
DeepSeek 67B	67	73.0
LLaMA 2 70B	70	69.0
Mixtral 8×7B	13	70.6
LLaMA 3 8B	8	65.0

★ Pareto-optimal: highest MMLU at fewest activated params

DeepSeek-V2 Chat (RL)

SFT (1.5M sessions) + GRPO RL → strongest open-source Chinese chat model



★ Outperforms all open-source and most closed-source models on AlignBench (Chinese)

Key Takeaways

Architectural Innovation Unlocks Efficient LLM Scaling

1. MLA: A Strictly Better KV-Cache Solution

Low-rank key-value joint compression outperforms MHA while cutting KV cache by 93.3% — superior to GQA/MQA tradeoffs.

2. DeepSeekMoE Enables Economical Training

Fine-grained experts with shared expert isolation achieve 42.5% lower training cost than a 67B dense model.

3. Top-Tier Performance at a Fraction of Activated Params

21B activated parameters match or exceed 70B-class dense models (LLaMA 3 70B, Qwen1.5 72B) on MMLU.

4. 5.76× Generation Throughput

Makes DeepSeek-V2 practical for large-scale deployment at real-world inference latencies.

5. Strong Multilingual Alignment

DeepSeek-V2 Chat (RL) is the strongest open-source chat model in Chinese and competitive with closed-source systems in English.

6. Open-Source Availability

Full model weights and architecture details released → enabling reproducible research and community improvements.